Lab: 01

Ashik_Thread.java

```
import java.util.concurrent.atomic.AtomicLong;

public class Ashik_Thread extends Thread {
    static AtomicLong safecounter = new AtomicLong(0);
    static long unsafecounter = 0;
    int increment;
    boolean threadsafe;
    long localcounter;

    public Ashik_Thread(int increments,
            boolean threadsafe) {

        this.increments = increments;
        this.threasafe = threadsafe;
    }
```

IT-24047

```
@Override
public void run(){
    for(int i=0 ; i<increments; i++){
    if(threadsafe){
        safecounter.incrementAndGet();
    }
    else{
        unsafecounter++;
    }

    localcounter++;
    }
}

public static void main(String[] args){
    throws InterruptedException{
```

IT-24047

```
if(args.length != 3){
System.out.println("Usage: java AshikThread <threads> <true/false>");
return;
}

int numberofthreads;
int increments;
boolean threadsafe;

try {
    numberofthreads = Integer.parseInt
        (args[0]);
    increments = Integer.parseInt(args[1]);
    if( !args[2].equalIgnoreCase("true");
```

IT-24047

```
catch (NumberFormatException e) {
System.out.println("threads and
increaments must be Integers!");
return;
}
```

IT-24047

```
if (numberOfthreads <=0 || increments<0)
    {
    system.out.println("InValid input!");
    return;
    }
safeCounter.set(0);
  unsafecounter =0;
Ashik_Thread[] threads = new
    Ashik_Thread[numberOf_Threads];
```

```
long startTime = System.nanoTime();
for (int i=0 ; i<numberofThread ; i++){
    thread[i] = new AshikThread (incrementer,
    threadsafe );
    thread[i].start();
}
```

IT-24047

```
System.out.println("\n==== Thread
    Counter Result ====");
System.out.println("Number of thread"+
    number of threads);
System.out.println(" Expected output!"+
    expected count);
System.out.println("Absolute Difference"+
    differenee);
```

```
System.out.println(" percentage Difference
    %.4f%%%n", percentageDifference);

System.out.printf("Execution Time: %.3f
    ms%n", (endTime - startTime) / 1_000_000.0
);
    }
}
```

IT-24047

output:

in pdf